

Mathematical Structure of the January Game Theory Simulation

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1 The Prisoner's Dilemma Payoff Structure

The simulation is built around the classical Prisoner's Dilemma (PD), defined by four payoff parameters:¹

- R (Reward): payoff for mutual cooperation
- S (Sucker's payoff): payoff for cooperating while the opponent defects
- T (Temptation): payoff for defecting while the opponent cooperates
- P (Punishment): payoff for mutual defection

These parameters satisfy the inequality

$$T > R > P > S,$$

which creates the dilemma: defection is individually optimal in a single interaction, even though mutual cooperation would yield a higher collective payoff.

A typical payoff matrix is

$$\begin{pmatrix} (R, R) & (S, T) \\ (T, S) & (P, P) \end{pmatrix}.$$

2 Why Defect Wins in a Static Tournament

In a static round-robin tournament, every strategy plays every other strategy a fixed number of times, and total payoff determines success. In this setting, *Defect* has a structural advantage:

- It receives T against unconditional cooperators.
- It never pays the cooperation cost associated with receiving S .
- Against other defectors, it consistently receives P .

Because no strategies are removed or adapted, the tournament rewards short-term payoff maximization. As a result, defection dominates even though it undermines cooperation in the long run.²

¹The Prisoner's Dilemma and its payoff structure are discussed extensively in the context of cooperation and strategic conflict in [1].

²This outcome mirrors the results of Axelrod's original computer tournaments, where defection performs well in purely static settings [1].

3 Replicator Dynamics

The evolutionary part of the simulation replaces fixed scores with population dynamics governed by the replicator equation:

$$\dot{x}_i = x_i (f_i - \bar{f}).$$

Here,

- x_i is the population share of strategy i ,
- f_i is the expected payoff (fitness) of strategy i ,
- $\bar{f} = \sum_j x_j f_j$ is the average population payoff,
- $\dot{x}_i$ is the time derivative of x_i .

Strategies that perform better than average grow in frequency, while those that perform worse shrink.³

4 Why Tit-for-Tat Becomes Dominant

Tit-for-Tat (TFT) cooperates initially and then mirrors its opponent's previous action. Under replicator dynamics, this conditional structure is crucial:

- TFT achieves sustained mutual cooperation with itself and other cooperative strategies, earning repeated R payoffs.
- It retaliates immediately against defectors, preventing long-term exploitation.
- Defectors can invade cooperators, but they cannot exploit TFT persistently.

As defectors spread, they reduce the population of cooperators they depend on, lowering their own average payoff. TFT, by contrast, maintains stable cooperation within its own cluster, making it dynamically stable under a wide range of parameters.⁴

5 Bifurcation in the Temptation Payoff

Varying the temptation payoff T changes the qualitative behavior of the system. For low values of T , cooperative strategies such as TFT form stable fixed points. As T increases past a critical threshold, these equilibria lose stability and defection can invade.

For sufficiently large T , strong retaliation restores stability, producing new cooperative fixed points. This bifurcation structure shows that cooperation is neither universally fragile nor guaranteed; its stability depends sensitively on the incentive to defect.⁵

³This formulation assumes a well-mixed population with random matching and payoff-proportional reproduction, a standard assumption in evolutionary game theory; see [3].

⁴The resistance of conditional cooperators like Tit-for-Tat to invasion is a central result in evolutionary game theory; see [1, 3].

⁵The analysis of stability and bifurcations in evolutionary games is treated rigorously in [2].

References

- [1] Robert Axelrod. *The Evolution of Cooperation*. Basic Books, New York, 1984.
- [2] Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge, 1998.
- [3] Martin A. Nowak. *Evolutionary Dynamics: Exploring the Equations of Life*. Harvard University Press, Cambridge, MA, 2006.